C291 - Spring 25 - System Programming with C & Unix

Assignment 6

1. Remove Duplicates (20 Marks)

Write a C program that takes a sorted integer array as input and removes duplicates in place (means modifying the original data structure directly without allocating additional memory) using pointers. Pointers make this implementation different and efficient because they allow you to directly manipulate memory addresses, enabling you to avoid creating a new array and minimizing the number of swaps or shifts required during duplicate removal. The program should print the array with duplicates removed.

Note: Test your function by calling it in main with different input arrays. Arrays can be hardcoded.

Implementation Details:

```
/*
        Function: printUniqueElements
        ---------------------------------
        prints unique elements from the pointer array

        Parameters:
        inputArray: Pointer to an integer array.
        lenArray: The length of the input array.

        Returns: void
        Note:
        - In a sorted array, all duplicate elements appear together.
        - This function prints the array with duplicates removed.
*/
```

void printUniqueElements (int *inputArray, int lenArray);

Example 1:

```
int inputArray[] = {1, 1, 2, 3, 3, 4, 4, 5, 5};
int len = sizeof(inputArray) / sizeof(inputArray[0]);

printUniqueElements(inputArray, len);
```

Result:
Unique Elements: 1 2 3 4 5

Example 2:

```
int inputArray[] = {2, 2, 3, 3, 3, 3, 4, 5, 5, 5};
int len = sizeof(inputArray) / sizeof(inputArray[0]);

printUniqueElements(inputArray, len);
```

Result:
Unique Elements: 2 3 4 5

This function printUniqueElements takes an array of sorted integers and its length as parameters. It prints the unique elements from the sorted array, with duplicate elements removed.

Note: The solution for this problem should be named as solution_1.c and it should be under C291-Spring-25/assignment6 where assignment6 would be the new directory you would create.

2. Reverse an Integer (20 marks)

Write a C program that reverses the digits of an integer using pointers. The program should take an integer input and use pointers to reverse its digits and print the reversed integer.

Implementation Details:

```
/*
        Function: reverseInteger
        --------------------------------
        Reverses the digits of an integer using pointers.

        Parameters:
        num: Pointer to the integer to be reversed.

        Returns: void
*/
```

void reverseInteger(int *num);

This function reverseInteger takes an integer, reverses its digits using pointers, and updates the original integer with the reversed value.

Example 1:

Input: 12345

Output: 54321

Example 2:

Input: 987654321

Output: 123456789

Note: The solution for this problem should be named as solution_2.c and it should be under C291-Spring-25/assignment6 where assignment6 would be the new directory you would create.

3. Perform Operations (20 marks)

Write a C program to implement calculator using pointers. You should ask the user two operands and what kind of arithmetic operation the user wants to run on them. This shouldn't be just a onetime operation, the user should be able to input the operands and the kind of operation which it wants to be performed on them as long as he wants to. Users may enter 'Q' to quit the program.

You need to write separate function for all the operations (+, -, *, /). In these functions you need to pass the variables as pointers and then perform the requested operation and return the result. To calculate the result, you should not use a third variable.

For example: Consider the function int sum(int *a, int *b); Here, the addition of two numbers must be stored in 'a' and returned.

Implementation Details:

```
/*
        Function: addition, subtraction, multiplication, division
        ----------------
        Perform operations on two integers using pointers.
         Parameters:
         a: Pointer to the first integer.
         b: Pointer to the second integer.
         Returns: void
*/
```

```
void addition(int *a, int*b);
void subtraction(int *a, int*b);
void division(int *a, int*b);
void multiplication(int *a, int*b);
```

Sample run:

Enter the operation you want to perform('+' for Addition,'-' for Subtraction,'*' for Multiplication, '/' for Division, Q to quit): +

Enter the operands for this operation : **7 8**

Addition of 7 and 8 = **15**

Enter the operation you want to perform('+' for Addition,'-' for Subtraction,'*' for Multiplication, '/' for Division, Q to quit): -

Enter the operands for this operation: **109 9**

Subtraction of 109 and 9 = **100**

Enter the operation you want to perform('+' for Addition,'-' for Subtraction,'*' for Multiplication, '/' for Division, Q to quit): **Q**

Goodbye!

Note: The solution for this problem should be named as solution_3.c and it should be under C291-Spring-25/assignment6 where assignment6 would be the new directory you would create.

4. Find Min and Max (20 marks)

Write a C program that takes an array of integers as input and finds the minimum and maximum elements within the array using a single for loop iteration (Find min and max element in the same for loop). The program should use pointer notation to iterate through the elements.

Note: You will be given 0 if you use individual for loops to calculate min and max. Test your function by calling it in main with different input arrays. Arrays can be hardcoded.

Implementation Details:

```
/*
        Function: findMinMax
        --------------------------------
        Find minimum and maximum element from an integer array using pointers.

        Parameters:
        inputArray: Pointer to an integer array
        lenArray: The length of the input array
        Min: Minimum element from the array
        Max: Maximum element from the array

        Returns: void
*/
```

void findMinMax(int *inputArray, int lenArray, int *min, int *max);

This function reverseInteger takes an integer, reverses its digits using pointers, and updates the original integer with the reversed value.

Sample Input:

```
int inputArray[] = {15, 56, 34, 78, 23};
int len = sizeof(inputArray) / sizeof(inputArray[0]);

findMinMax(inputArray, len, min, max);
```

Result:
Minimum element: 15
Maximum element: 78

Note: The solution for this problem should be named as solution_4.c and it should be under C291-Spring-25/assignment6 where assignment6 would be the new directory you would create.

5. Copy one array to another (20 marks)

Write a C program that takes a hardcoded array of integers and copy the elements of this hardcoded array into another array using pointers and use pointer notation to iterate through the elements. Displays the copied array as the output. You are required to implement the function **copyArray** and call that function from the main method.

Note: Test your function by calling it in main with different input arrays. Arrays can be hardcoded.

Implementation Details:

```
/*
        Function: CopyArray
        --------------------------------
        Copy source array to destination array.

        Parameters:
        srcArray: Pointer to a source array
        destArray: Pointer to a destination array
        lenArray: Length of the source array

        Returns: void
*/
```

void copyArray(int *srcArray, int *destArray, int lenArray);

This function copyArray takes an integer array as input (Source Array) and copy source array to Destination Array.

Sample Input:

```
int srcArray[] = {15, 56, 34, 78, 23};
int lenArray = sizeof(srcArray) / sizeof(srcArray[0]);
int destArray[lenArray];
copyArray(srcArray, destArray, lenArray);
```

Result:
Destination Array: 15, 56, 34, 78, 23 (Make sure you print the commas also)

Note: The solution for this problem should be named as solution_5.c and it should be under C291-Spring-25/assignment6 where assignment6 would be the new directory you would create.